Assignment 2 part 3

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The code can be found here: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_2_3
The file is called 'FOR_GRADING_A2_P3'

In this part of the assignment, we try to figure out how a new study should be planned (i.e. how many participants are needed) in order to have enough power to replicate the findings (ensuring our sample size is adequate, our alpha at 0.05 and our beta at 0.8):

Introduction

We will be working with two datasets: dataset A, which entails a full dataset consisting of 346 rows (66 participants), and dataset B, which functions as a "pilot" dataset consisting of 36 rows (6 participants). In the preprocessing of the data, 26 rows were excluded in dataset A by criteria of missing values in the variables "CHI_MLU" (child MLU) and "age".

All power analyses were conducted with the following model using the package lme4 (Bates, Maechler, Bolker & Walker, 2015).

model1 = child_MLU ~ Diagnosis * Visit + (Visit|Subject)

From the previous parts of this assignment we know that a significant effect of the interaction between Diagnosis and Visit can be found. Thus, we will only examine the interaction effect and thereby disregard the main effects when conducting the following power analyses.

All power analyses were conducted by running 50 simulations using the package simr (Green & MacLeod, 2016) to test the sufficient number of participants needed to get a power of at least 80% at an alpha level of 0.05.

1. if we trust the estimates of the current study. Report the power analysis and comment on what you can (or cannot) use its estimates for.

A power analysis on dataset A using model1 showed that the interaction Diagnosis:Visit had a power of 100 %, CI 95 % [93, 100]. In figure 1 it can be seen that 25 participants are needed to obtain sufficient power of 80%. The current study has 66 participants. We can thereby conclude that the current study was overpowered.

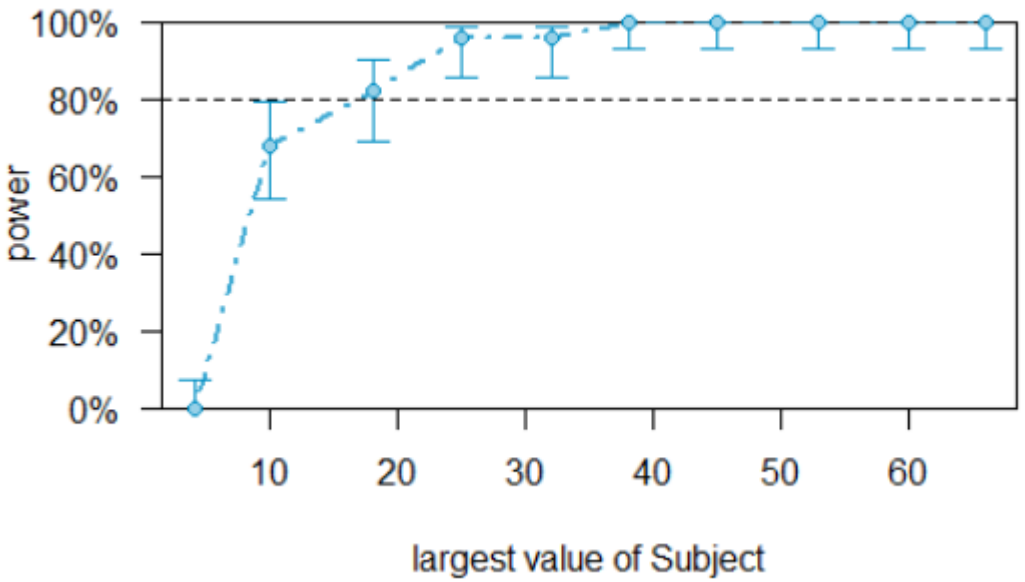

Figure 1: Power plot on dataset A. The effect size of Diagnosis:Visit is 0.25. 25 participants are needed to obtain a power of >80%.

The estimate of Diagnosis:Visit is 0.25. Thus, typically developing children will have a mean length of utterance that is 1.5 syllables longer than children with an autism spectrum disorder at visit 6. We argue that this estimate is too high, and would not necessarily be replicated in a follow-up study. Therefore, we lower the estimate for a more conservative power analysis.

2. if we are skeptical of the current study. Report the power analysis and comment on what you can (or cannot) use its estimates for.

Full study

In order to run a more conservative power analysis the effect size can be set to a lower value. However, the effect size should still be theoretical justifiable. We have decided that a difference of minimum 1 syllable per utterance from visit 1 to visit 6 is sufficient for a meaningful effect size. Thus, the lowest justifiable effect size would be 1 syllable divided by 6 visits, which equals 0.167.

A power analysis was conducted on model 1 by simulating dataset A at an effect size of 0.167 (see the section ‘Introduction’ for details) . Results showed that a total sample of 53 participants was required to achieve a power of 94% with CI 95% [83.45, 98.75] (figure 2).

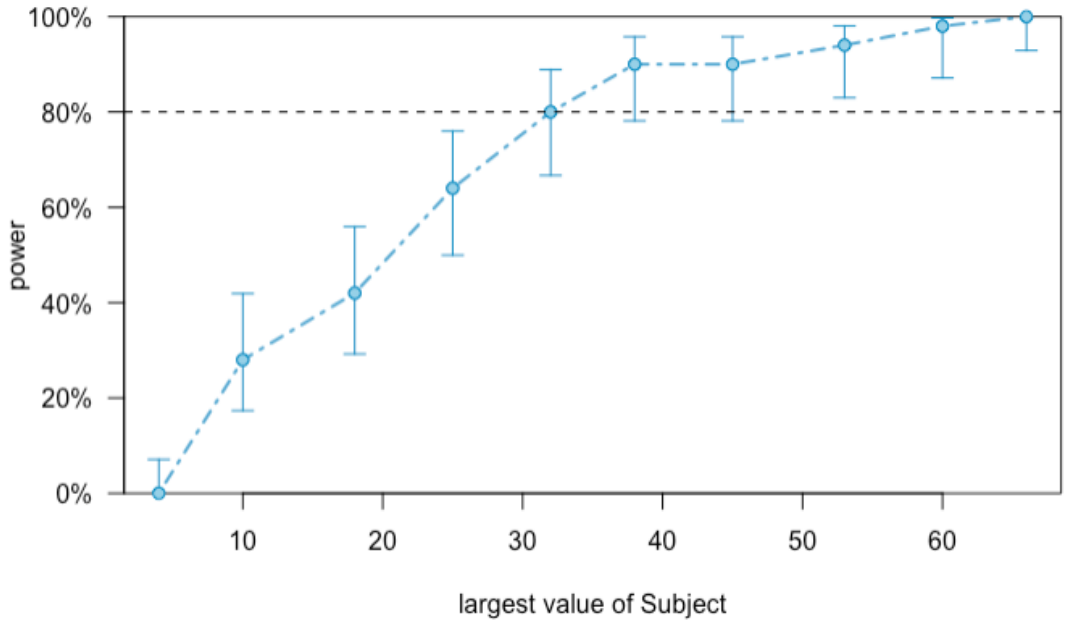

Figure 2: Power curve for an effect size of 0.167 simulating dataset A on model 1. 53 participants are sufficient to get a power of >80%.

Pilot study

Since we had access to another dataset (dataset B) we decided to analyse it as it was a pilot study with 6 participants. Note that this is not a legit pilot-study, but merely an exploratory exercise to investigate the validity of a power analysis on a pilot study.

First we conducted a power analysis on dataset B. We found that with 6 participants and an effect size of 0.167 the pilot study had a power of 10 % with CI 95% [3.33, 21.81]. We then conducted a power analysis where the number of participants had been extended, to investigate the sufficient number of participants needed to obtain a power of >80%. We found that a power of 92% with CI 95% [80.77,97.78] were reached with 48 participants (figure 3). This estimate was a good approximation of the estimate of 53 participants in our analysis using dataset A. However, it is important to take into account that it is assumed that the pilot study data is representative of the population, which might be violated since dataset B only had 6 participants, that is n = 3 in each group (ASD, TD).
All in all this experiment supports the notion that extending the number of subjects from a pilot study gives a good approximation of the number of participants needed to obtain a power of >80%.

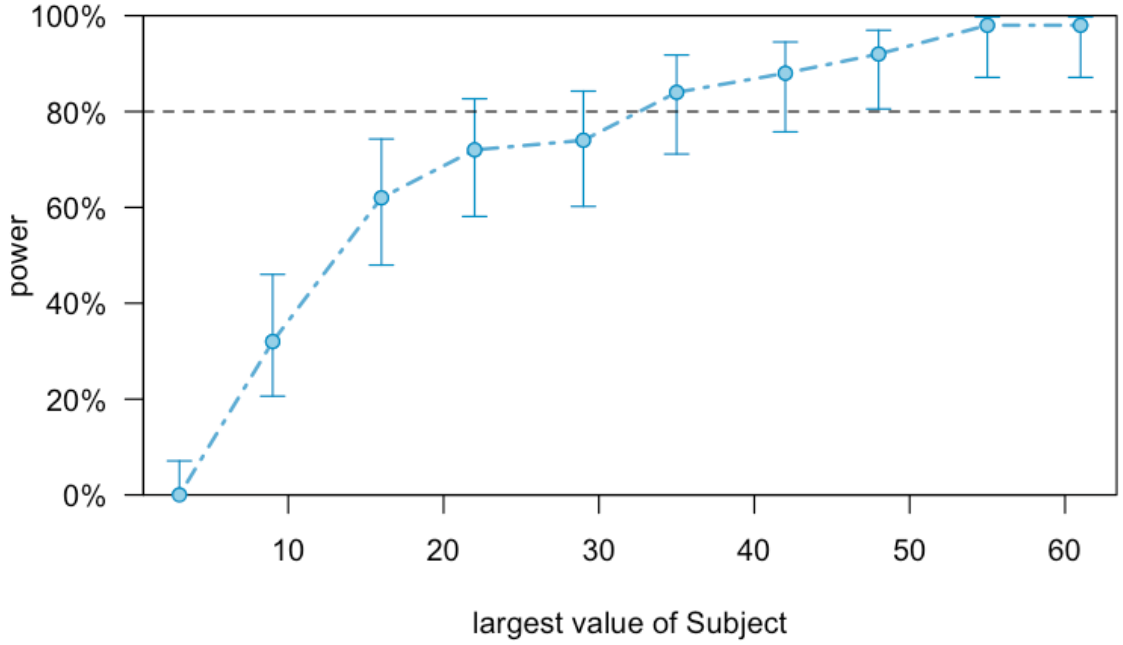

Figure 3: Power curve for an effect size of 0.167 simulating dataset B on model 1 as a pilot study. 48 participants are sufficient to get a power of >80%.

3. if we only have access to 30 participants. Identify the power for each relevant effect and discuss whether it's worth to run the study and why

We assessed the power for the interaction effect between Diagnosis and Visit with n = 30 using the function extend(n=30). We found that with 30 participants and an effect size of 0.167, the study had a power of 86 % with CI 95% [73.26 , 94.18]. The obtained power of 30 participants is insufficient because of the large confidence interval that drops below 80%.
It is therefore not worth it to run the study with only 30 participants.

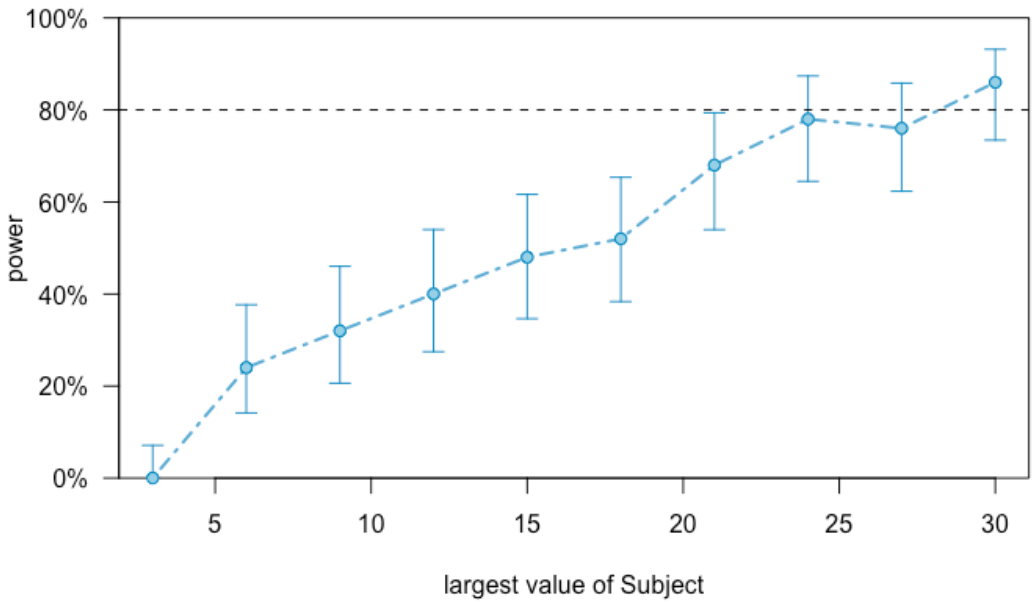

Figure 4: Power curve for the dataset with 30 participants.

Extending visit

We cannot recruit more participants to gain more power, but we can extend the number of visits to gain more power. Figure 5 shows the result of extending to 10 visits and it shows that 7 visits with 30 participants would give a power of 92.00% with CI = [80.77, 97.78].

However, we know from the previous parts of the assignment that the effect of MLU is best described by a growth curve model rather than a linear model. This means that the effect will stagnate along visits. Model 1 uses the linear method, why extending visit will give an inaccurate estimate of power for the true effect.

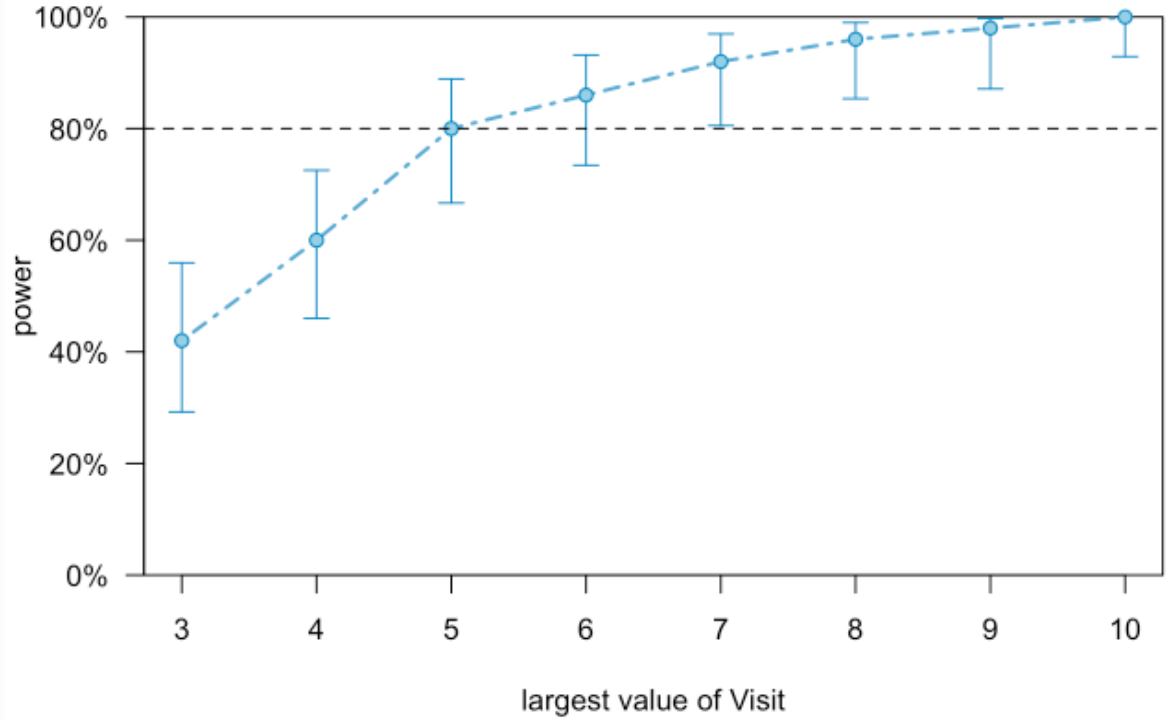

Figure 5: Power curve for an effect size of 0.167 simulating dataset A on model 1 with only 30 participants. Extending visit to 7 is sufficient to get a power of >80% with only 30 participants.

References

- Bates, Maechler, Bolker & Walker (2015). Fitting Linear Mixed-Effects Models Using lme4. Journal of Statistical Software, 67(1), 1-48. doi:10.18637/jss.v067.i01.

- Green & MacLeod (2016). simr: an R package for power analysis of generalised linear mixed models by simulation. Methods in Ecology and Evolution, 7(4), 493-498. doi: 10.1111/2041-210X.12504 (URL: https://doi.org/10.1111/2041-210X.12504), <URL: https://CRAN.R-project.org/package=simr>.